

D599

Data Preparation and Exploration

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A clinic's patient-income column is strongly right-skewed, with about 4% of values missing at random. The data covers 18 months and was entered by three front-desk clerks. The analyst wants to fill the gaps without shifting the center of the distribution. Which imputation is most appropriate?

- A. Fill missing values with 0
- B. Median imputation
- C. Delete the column, since any missing data makes it unusable
- D. Mean imputation

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

Delivery times for a courier have $Q1 = 20$ minutes, $Q3 = 40$ minutes, and a median of 28 minutes. The courier covers 6 zip codes and has 45 drivers. Using the $1.5 \times IQR$ rule, what is the upper fence for outliers?

- A. 58 minutes
- B. 60 minutes
- C. 70 minutes
- D. -10 minutes

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A dataset's upper outlier fence under the $1.5 \times IQR$ rule is 130, and $Q3$ is 100. What is $Q1$?

- A. 70
- B. 80
- C. 20
- D. 120

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A real-estate brokerage with offices in four suburbs reports this year's sale prices: most homes sold between \$300,000 and \$450,000, plus three estate sales over \$4 million. The brokerage won a regional award last year. Which measure best represents a typical sale price?

- A. The standard deviation
- B. The midrange, halfway between lowest and highest price
- C. The median
- D. The mean

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A city transit agency surveyed 600 of its 90,000 riders; 62% favored later weekend service. The survey ran during a rainy week in March. The agency wants to state a likely range for support among all riders. Which approach fits?

- A. Report 62% as the exact figure for all 90,000 riders.
- B. Build a confidence interval for the population proportion from the sample.
- C. Survey all 90,000 riders, since a sample can never describe a population.
- D. Adjust the result for the rainy weather first.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A product team at an online retailer with 1.2 million monthly visitors runs an A/B test on a new checkout button. The new button is green. With $\alpha = 0.05$, the test returns $p = 0.03$. What is the correct conclusion?

- A. The test proves the new button raises conversions.
- B. Fail to reject the null hypothesis.
- C. Reject the null hypothesis; the data show a statistically significant difference.
- D. There is a 3% chance the null hypothesis is true.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

An analyst's test returns $p = 0.08$ at $\alpha = 0.05$. What is the correct conclusion?

- A. Fail to reject the null; the evidence is not strong enough at this α .
- B. The null hypothesis is proven true.
- C. There is a 92% chance the alternative is true.
- D. Reject the null.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A hospital group compares mean length of stay across three of its hospitals (built in 1978, 1995, and 2012). Length of stay is roughly normal in each hospital. Which test fits?

- A. One-way ANOVA
- B. Chi-square test of independence
- C. A regression of stay length on the year each hospital was built
- D. Independent two-sample t-test

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

An analyst measures the same 40 employees' productivity before and after a training program. Which test fits?

- A. One-way ANOVA
- B. Chi-square test
- C. Independent two-sample t-test
- D. Paired t-test

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A bank's fraud team, which has 22 investigators, tests the null hypothesis that a given transaction is legitimate. The bank's fraud losses were \$3 million last year. In this setting, what is a Type I error?

- A. Losing \$3 million to fraud.
- B. Letting a fraudulent transaction pass as legitimate.
- C. An error that disappears entirely if α is set to 0.
- D. Flagging a legitimate transaction as fraud.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A team lowers α from 0.05 to 0.01 and keeps the same sample size. What happens to the risk of a Type II error?

- A.** It increases.
- B.** It is unchanged, since α affects only Type I error.
- C.** It decreases.
- D.** Both error risks fall to zero.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

An analyst at a 300-employee insurer wants to show whether agents' years of experience relate to the number of policies they sold. The chart will appear in the company's blue-and-gold report template. Which chart fits?

- A.** Pie chart of policies by agent
- B.** Histogram of policies sold
- C.** Bar chart of policies by agent name
- D.** Scatter plot of experience vs. policies sold

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A clinic's patient-income column is strongly right-skewed, with about 4% of values missing at random. ~~The data covers 18 months and was entered by three front desk clerks.~~ The analyst wants to fill the gaps without shifting the center of the distribution. Which imputation is most appropriate?

Target word: imputation method **Decides:** Right-skewed variable, small share missing at random.

A	Confusion set	Zero is a real income value, not a stand-in for unknown.
B	Correct	The median is not pulled by the long right tail, so the center is preserved.
C	Absolute language	A 4% gap is routinely handled; dropping the variable loses information.
D	Adjacent category	The skew pulls the mean up, so filled values overstate typical income.

2. Answer C Concept 2

Delivery times for a courier have $Q1 = 20$ minutes, $Q3 = 40$ minutes, and a median of 28 minutes. ~~The courier covers 6 zip codes and has 45 drivers.~~ Using the $1.5 \times IQR$ rule, what is the upper fence for outliers?

Target word: upper fence **Decides:** $IQR = 20$; $1.5 \times 20 = 30$ added to $Q3$.

A	Wrong stem number	Adds 30 to the median instead of $Q3$.
B	Dropped element	Adds the IQR without the 1.5 multiplier.
C	Correct	$40 + 1.5 \times (40 - 20) = 40 + 30 = 70$.
D	Inverted operation	That is the lower fence, $Q1 - 30$.

3. Answer B Concept 2 · transfer

A dataset's upper outlier fence under the $1.5 \times IQR$ rule is 130, and $Q3$ is 100. What is $Q1$?

Target word: solve backward for $Q1$ **Decides:** $1.5 \times IQR = 30$, so $IQR = 20$.

A	Dropped element	Treats 30 as the IQR, forgetting to divide by 1.5.
B	Correct	$130 - 100 = 30 = 1.5 \times IQR \rightarrow IQR = 20 \rightarrow Q1 = 100 - 20 = 80$.
C	Adjacent category	That is the IQR, not $Q1$.
D	Inverted operation	Adds the IQR to $Q3$ instead of subtracting.

4. Answer C Concept 3

A real-estate brokerage ~~with offices in four suburbs~~ reports this year's sale prices: most homes sold between \$300,000 and \$450,000, plus three estate sales over \$4 million. ~~The brokerage won a regional award last year.~~ Which measure best represents a typical sale price?

Target word: best measure of center **Decides:** A few extreme high values skew the data.

A	Confusion set	Measures spread, not a typical value.
B	Dropped element	Uses only two values, one of them an extreme.
C	Correct	The middle value is not pulled by the three estate sales.
D	Adjacent category	The estate sales drag the mean well above a typical sale.

5. Answer B Concept 4

A city transit agency surveyed 600 of its 90,000 riders; 62% favored later weekend service. ~~The survey ran during a rainy week in March.~~ The agency wants to state a likely range for support among all riders. Which approach fits?

Target word: descriptive or inferential **Decides:** A sample is used to speak for the whole population.

A	Dropped element	Ignores sampling error; that is a descriptive number for the sample only.
B	Correct	Inference estimates the population value with a margin of error.
C	Absolute language	Sound samples are the basis of inference.
D	Camouflage	The weather detail does not decide the method.

6. Answer C Concept 5

A product team at an online retailer ~~with 1.2 million monthly visitors~~ runs an A/B test on a new checkout button. ~~The new button is green.~~ With $\alpha = 0.05$, the test returns $p = 0.03$. What is the correct conclusion?

Target word: decision from p and α **Decides:** $p = 0.03 \leq 0.05$.

A	Absolute language	Tests give evidence, not proof.
B	Inverted operation	That is the decision when $p > \alpha$.
C	Correct	p is at or below α , so the null is rejected.
D	Confusion set	p is the chance of data this extreme if the null were true, not the chance the null is true.

7. Answer A Concept 5 · transfer

An analyst's test returns $p = 0.08$ at $\alpha = 0.05$. What is the correct conclusion?

Target word: decision when p exceeds α **Decides:** $0.08 > 0.05$.

A	Correct	p above α means the null is not rejected.
B	Absolute language	Failing to reject is not proof of no effect.
C	Confusion set	$1 - p$ is not the probability of the alternative.
D	Inverted operation	Rejection requires $p \leq \alpha$.

8. Answer A Concept 6

A hospital group compares mean length of stay across three of its hospitals ~~(built in 1978, 1995, and 2012).~~ Length of stay is roughly normal in each hospital. Which test fits?

Target word: which parametric test **Decides:** Three independent group means.

A	Correct	Compares means across three or more groups in one test.
B	Confusion set	For two categorical variables, not a numeric mean.
C	Camouflage	Answers the build-year detail rather than comparing the three means.
D	Adjacent category	Handles only two groups.

9. Answer D Concept 6 · transfer

An analyst measures the same 40 employees' productivity before and after a training program. Which test fits?

Target word: which test for repeated measures **Decides:** The same people are measured twice.

A	Adjacent category	For three or more independent groups.
B	Dropped element	Drops the fact that productivity is numeric.
C	Confusion set	Assumes two separate groups of people.
D	Correct	Compares before and after differences within the same subjects.

10. Answer D Concept 7

A bank's fraud team, ~~which has 22 investigators,~~ tests the null hypothesis that a given transaction is legitimate. ~~The bank's fraud losses were \$3 million last year.~~ In this setting, what is a Type I error?

Target word: meaning of Type I error here **Decides:** Null = legitimate; Type I = rejecting a true null.

A	Camouflage	The loss figure does not define the error type.
B	Inverted operation	That is a Type II error, a miss.
C	Absolute language	$\alpha = 0$ means never rejecting, which maximizes Type II error.
D	Correct	Rejects a true null: a false alarm.

11. Answer A Concept 7 · transfer

A team lowers α from 0.05 to 0.01 and keeps the same sample size. What happens to the risk of a Type II error?

Target word: effect on the other error **Decides:** A stricter α makes rejection harder.

A	Correct	Harder to reject means more real effects are missed.
B	Dropped element	The two errors trade off at a fixed sample size.
C	Inverted operation	Lower α reduces Type I risk, not Type II.
D	Absolute language	Neither error can be eliminated.

12. Answer D Concept 8

An analyst ~~at a 300-employee insurer~~ wants to show whether agents' years of experience relate to the number of policies they sold. ~~The chart will appear in the company's blue and gold report template.~~ Which chart fits?

Target word: chart for a relationship **Decides:** Two numeric variables and the question "relate to."

A	Confusion set	Shows parts of a whole, not a relationship.
B	Dropped element	Shows one variable's distribution and drops experience.
C	Adjacent category	Compares agents but does not show experience against sales.
D	Correct	Shows the relationship between two numeric variables.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Imputing missing values	Choose imputation that preserves the variable's shape: median for skewed data, mean only for roughly symmetric data; never fill with a value that has real meaning (such as 0).	Filling with the mean or zero by habit, or deleting whole columns for a few missing cells.	A skewed column with a small share missing.
2	IQR outlier fences	$IQR = Q3 - Q1$; lower fence = $Q1 - 1.5 \times IQR$; upper fence = $Q3 + 1.5 \times IQR$.	Building fences from the median, or forgetting the 1.5 multiplier.	Q1, Q3, and "1.5 × IQR" in the stem.
3	Center in skewed data	The median resists extreme values; the mean is pulled toward the tail.	Reporting the mean as the typical value when a few extreme values exist.	A handful of values far larger than the rest.
4	Descriptive vs. inferential statistics	Descriptive statistics summarize the data in hand; inferential statistics use a sample to estimate a population with a measure of uncertainty.	Reporting a sample percentage as if it were the exact population value.	A sample of a larger group and a wish to speak for the whole group.
5	Reading a hypothesis test	If $p \leq \alpha$, reject the null hypothesis; if $p > \alpha$, fail to reject it. A test never proves either hypothesis true.	Saying the test "proves" the effect, "accepts" the null, or that p is the probability the null is true.	A p-value next to an alpha level.
6	Choosing a parametric test	Two independent group means → independent t-test; same subjects measured twice → paired t-test; three or more group means → ANOVA; two categorical variables → chi-square.	Using a t-test for three groups or an independent test for before-and-after data.	How many groups, whether subjects repeat, and whether the outcome is numeric or categorical.
7	Type I and Type II errors	Type I = rejecting a true null (false alarm); Type II = failing to reject a false null (miss). Lowering α cuts Type I risk but raises Type II risk at a fixed sample size.	Swapping the two, or thinking both can be driven to zero.	The null stated in business words plus "false alarm" or "missed."
8	Matching the chart to the question	Scatter = relationship between two numeric variables; histogram = one variable's distribution; bar = comparing categories; line = change over time.	Picking a familiar chart that cannot show the relationship asked about.	"Relates to," "distribution," "compare," or "over time."

Course scope (public description): Data Preparation and Exploration applies analytical programming skills to the early steps of the data analytics life cycle. This course covers cleaning data to ensure the structure, accuracy, and quality of the data; interpretation of descriptive and inferential statistics as well as visualizations of data; and wrangling data to prepare it for further analysis. The course introduces hypothesis testing, focusing on application for parametric tests, and addresses communication skills and tools to explain an analyst's findings to others within an organization. The following courses are prerequisites: The Data Analytics Journey, Data Management, and

Analytics Programming.